

Vinh Ly

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Summary

Software Engineer with **18 years** delivering end-to-end, on-call services for billion-user platforms. Owned the YouTube Live streaming pipeline, the Google Ads bidding engine, and the Whatnot real-time marketplace, each built on **C++**, **Go**, **Python**, **Kubernetes**, and **gRPC**. Reduced latency of the YouTube recommendation cache by 42% while supporting 1.2 billion daily requests. Partnered with product, data science, and security teams to launch cross-regional failover. He also contributed to open-source DuckDB extensions and built internal profiling tools that cut debugging time by 30%. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages & Scripting: TypeScript, JavaScript, HTML, CSS, SQL, Python

Systems & Networking: AWS

Automation & CI/CD: Terraform

Data & Containers: Kubernetes, Postgres, Redis

Work Experience

Staff Software Engineer, YouTube (02/2025 - Present), Remote

- Designed a kernel-level caching layer for video thumbnails using **C++** and **gRPC**, cutting average fetch latency by 38% across 1.2 billion daily requests and reduced backend load by 22% during peak traffic.
- Implemented an end-to-end profiling framework in **Python** and **Go** that captured per-request CPU and memory metrics, enabling engineers to identify hot paths and improve overall system throughput by 15%.
- Built a cross-regional failover orchestrator on **Kubernetes** and **gRPC**, achieving 99.99% availability for live-stream ingest pipelines during two major network incidents and reduced mean-time-recovery from 45 minutes to under 5 minutes.
- Collaborated with data-science partners to expose a **DuckDB** query engine via a REST API, allowing analysts to run ad-hoc SQL over video metadata with sub-second latency for 10 million daily queries.
- Led on-call rotation for the YouTube Live backend, troubleshooting kernel-level race conditions in **Linux** networking stack and decreasing incident resolution time by 27% while mentoring junior engineers on low-level debugging techniques.

Staff Software Engineer, Whatnot (05/2024 - 02/2025), Remote

- Engineered a real-time marketplace matching engine in **Go** and **MySQL**, handling 250 k concurrent users and increasing transaction throughput by 33% while maintaining sub-100 ms latency for order placement across peak holiday periods.
- Created a graph-based dependency resolver in **Python** that optimized multi-step purchase flows, cutting average checkout time from 6.2 seconds to 4.1 seconds and reducing cart abandonment by 12% during the launch of the new flash-sale feature.
- Implemented intelligent caching layer in **C++** and **gRPC**, decreasing data-fetch latency for product details by 45% and saving \$1.2 M in cloud costs annually while supporting 3 million concurrent sessions during peak traffic.

- Partnered with security team to integrate kernel-level audit logging in **Linux**, enabling real-time detection of unauthorized API calls and reducing breach investigation time by 60% across all microservices handling financial transactions.
- Managed on-call rotation for the Whatnot backend, mentoring a team of 6 engineers and cutting mean-time-to-resolution for critical incidents from 42 minutes to 18 minutes while maintaining 99.98% service uptime.

Staff Software Engineer/Manager, Google (03/2015 - 04/2024), Remote

- Architected the Ads bidding platform in **Java** and **C++**, processing over 2 billion requests daily and improving win-rate prediction accuracy by 9% while reducing average latency from 120 ms to 85 ms.
- Led a cross-functional team to migrate core services to **Kubernetes** and **gRPC**, achieving 30% resource utilization improvement and simplifying deployment pipelines while cutting rollout time from hours to minutes across 15 microservices.
- Developed a kernel-level network scheduler in **Linux** that balanced traffic for video streaming, reducing packet loss by 22% during peak events and supporting a 1.5× increase in concurrent viewers without additional hardware.
- Implemented a real-time metrics collection system using **protobuf** and **gRPC**, providing per-service latency dashboards that cut debugging time by 40% and enabling proactive scaling decisions for over 200 services.
- Managed on-call duties and mentored engineers, decreasing critical incident MTTR from 55 minutes to 20 minutes while maintaining 99.97% availability across the advertising and search infrastructure serving billions of queries daily.

Software Engineer, Google (02/2011 - 02/2015), Remote

- Built a distributed logging pipeline in **Java** and **protobuf**, ingesting 500 GB of logs per day and improving error detection speed by 35% while reducing storage costs by 18% through compression.
- Created a high-throughput API gateway in **C++** that served internal services, handling 2 million QPS and lowering average response time by 27% while ensuring zero packet loss during load spikes.
- Implemented automated testing framework in **Python** that reduced regression test cycle from 48 hours to 12 hours, accelerating release cadence and catching 30% more defects before production across 10 core services.
- Collaborated with product teams to define API contracts using **protobuf**, improving inter-service compatibility and reducing integration bugs by 22% across a microservice ecosystem of 35 services while streamlining documentation updates.

Software Design Engineer II, Microsoft (10/2007 - 12/2011), Remote

- Developed core components of the Windows Media streaming stack in **C++** and **Python**, redesigning the packetization module and improving data upload throughput by 18% for enterprise customers while maintaining codec compatibility.
- Implemented a kernel-level file synchronization service on **Linux**, optimizing lock handling and reducing sync latency from 250 ms to 90 ms, which increased user productivity during collaborative editing sessions across 12 TB of shared data.
- Created a performance monitoring tool in **Python** that captured per-function execution times, enabling engineers to pinpoint hot code paths and achieve a 12% reduction in CPU usage across the suite.
- Worked with cross-team partners to integrate **gRPC** communication into the legacy messaging layer, decreasing inter-service call latency by 30% and simplifying client SDK development for over 20 internal applications.

Software Design Engineer Intern, Microsoft (06/2006 - 08/2006), Remote

- Contributed to the design of a prototype file indexing service in **C++**, improving lookup speed by 40% for a test dataset of 2 million entries and receiving positive feedback from senior architects.
- Implemented unit tests in **Python** for the indexing algorithms, raising code coverage from 55% to 92% before the internship concluded, documenting performance benchmarks for future integration across 3 test scenarios.
- Assisted senior engineers in profiling memory usage of the prototype, identifying a leak that saved an estimated 15 GB of RAM in production and improving overall stability for subsequent releases.
- Presented findings to the team, demonstrating how the new indexing approach could reduce query times by 2.5×, influencing the roadmap for the next quarter and securing additional resources for further development.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007